

D415

Software Defined Networking

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional bank, which has 60 branches, deploys an SDN controller that computes paths for every switch. The switches then forward customer traffic at line rate. The IT director presented the plan at a conference. Which plane do the switches mainly perform?

- A. The control plane.
- B. The management plane only.
- C. The data plane.
- D. Both planes equally, as before SDN.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In an SDN network, which component decides the path a new flow should take?

- A. Each switch's forwarding table on its own.
- B. The underlay cabling.
- C. The controller.
- D. The end host.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A developer, who learned to code at a weekend bootcamp, writes a monitoring app that asks the SDN controller for a list of devices through a REST API. Which interface is this?

- A. OpenFlow.
- B. Southbound.
- C. East-west.
- D. Northbound.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

The controller pushes flow entries into switches using OpenFlow. Which interface is this?

- A. Console cable.
- B. REST API.
- C. Northbound.
- D. Southbound.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A hospital network team enters the goal “Medical imaging traffic must always get priority and stay isolated from guest Wi-Fi.” The system turns this into device configs and alerts the team if the network drifts from it. The team moved into a new office last month. What approach is this?

- A. A one-time script.
- B. Network address translation.
- C. Manual CLI configuration.
- D. Intent-based networking.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A retailer runs its whole SDN on one controller in its main data center, which has excellent cooling. What is the biggest design risk?

- A. The controller is a single point of failure; it should be clustered.
- B. Cooling may fail.
- C. The switches will stop forwarding data-plane traffic by design.
- D. There is no risk; centralization always improves reliability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A cloud provider builds virtual networks for each customer on top of its physical IP fabric by wrapping customer traffic in VXLAN headers. The provider has data centers on three continents. What are the virtual networks?

- A. VLAN trunks.
- B. Overlays.
- C. A replacement for the physical network.
- D. The underlay.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An engineer at a university, who is also a part-time instructor, writes a file stating “VLAN 30 exists on all access switches, named Labs.” The tool figures out what to change on each switch. What style of automation is this?

- A. Declarative only if written in Python.
- B. Declarative.
- C. Imperative.
- D. Manual.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which is an example of imperative automation?

- A. A script that logs in to each switch, enters config mode, and types each VLAN command in order.
- B. A monitoring dashboard.
- C. An intent such as “isolate guest traffic.”
- D. A file listing the VLANs that should exist.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An automation playbook that adds an NTP server to 200 routers is run twice by mistake. The playbook was written during a holiday week. If it is idempotent, what happens on the second run?

- A. No changes are made, since the NTP server is already configured.
- B. The routers reboot.
- C. Every rerun always fails.
- D. Each router gets a duplicate NTP entry.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A data center, built on a former farm site, uses 4 spine switches and 20 leaf switches in a spine-leaf design. How many spine-to-leaf links are needed?

- A.** 24
- B.** 80
- C.** 20
- D.** 190

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A spine-leaf fabric has 96 spine-to-leaf links and 6 spines. How many leaves are there?

- A.** 90
- B.** 576
- C.** 16
- D.** 96

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A regional bank, ~~which has 60 branches,~~ deploys an SDN controller that computes paths for every switch. The switches then forward customer traffic at line rate. ~~The IT director presented the plan at a conference.~~ Which plane do the switches mainly perform?

Target word: role of switches in SDN **Decides:** Switches forward; the controller decides.

A	Inverted operation	The controller holds the control plane here.
B	Adjacent category	Management covers configuration access, not forwarding.
C	Correct	Forwarding traffic is the data plane.
D	Dropped element	Drops the separation SDN introduces.

2. Answer C Concept 1 · transfer

In an SDN network, which component decides the path a new flow should take?

Target word: who decides paths **Decides:** Path decisions are control-plane work.

A	Inverted operation	Tables carry out decisions the controller installs.
B	Adjacent category	Physical links carry traffic but make no decisions.
C	Correct	Centralized control plane.
D	Confusion set	Hosts do not choose network paths.

3. Answer D Concept 2

A developer, ~~who learned to code at a weekend bootcamp,~~ writes a monitoring app that asks the SDN controller for a list of devices through a REST API. Which interface is this?

Target word: interface direction **Decides:** App to controller.

A	Confusion set	A southbound protocol.
B	Inverted operation	Controller-to-device.
C	Adjacent category	Controller-to-controller.
D	Correct	Apps reach the controller northbound.

4. Answer D Concept 2 · transfer

The controller pushes flow entries into switches using OpenFlow. Which interface is this?

Target word: interface direction **Decides:** Controller to devices.

A	Adjacent category	Manual management access.
B	Confusion set	Typically used northbound.
C	Inverted operation	App to controller.
D	Correct	Controller programming devices.

5. Answer D Concept 3

A hospital network team enters the goal “Medical imaging traffic must always get priority and stay isolated from guest Wi-Fi.” The system turns this into device configs and alerts the team if the network drifts from it. ~~The team moved into a new office last month.~~ What approach is this?

Target word: approach **Decides:** Business goal translated and continuously verified.

A	Dropped element	Drops the continuous verification.
B	Confusion set	Unrelated address mapping.
C	Inverted operation	The team did not write device commands.
D	Correct	Intent, translation, and assurance.

6. Answer A Concept 4

A retailer runs its whole SDN on one controller in its main data center, ~~which has excellent cooling.~~ What is the biggest design risk?

Target word: design risk **Decides:** One controller.

A	Correct	Redundancy protects the control plane.
B	Camouflage	The stem says cooling is excellent; not the key design flaw.
C	Adjacent category	Existing flows may keep forwarding; the risk is losing control.
D	Absolute language	Centralization concentrates failure risk.

7. Answer B Concept 5

A cloud provider builds virtual networks for each customer on top of its physical IP fabric by wrapping customer traffic in VXLAN headers. ~~The provider has data centers on three continents.~~ What are the virtual networks?

Target word: term **Decides:** Virtual networks built on top of the physical network.

A	Confusion set	Layer 2 tagging, not IP-based encapsulation.
B	Correct	Encapsulated virtual networks.
C	Absolute language	Overlays depend on the underlay.
D	Inverted operation	The physical fabric is the underlay.

8. Answer B Concept 6

An engineer at a university, ~~who is also a part-time instructor,~~ writes a file stating “VLAN 30 exists on all access switches, named Labs.” The tool figures out what to change on each switch. What style of automation is this?

Target word: style **Decides:** States the desired end state only.

A	Confusion set	Style is about approach, not language.
B	Correct	Describes what, not how.
C	Inverted operation	Imperative lists steps.
D	Adjacent category	A tool made the changes.

9. Answer A Concept 6 • transfer

Which is an example of imperative automation?

Target word: pick imperative **Decides:** Explicit ordered steps.

A	Correct	Explicit steps.
B	Confusion set	Not automation of changes.
C	Adjacent category	Intent-based.
D	Inverted operation	Declarative.

10. Answer A Concept 7

An automation playbook that adds an NTP server to 200 routers is run twice by mistake. ~~The playbook was written during a holiday week.~~ If it is idempotent, what happens on the second run?

Target word: effect of idempotent rerun **Decides:** State already matches.

A	Correct	Idempotent tasks change only what differs.
B	Adjacent category	Not implied.
C	Absolute language	Idempotent reruns succeed with no change.
D	Inverted operation	Describes a non-idempotent task.

11. Answer B Concept 8

A data center, ~~built on a former farm site,~~ uses 4 spine switches and 20 leaf switches in a spine-leaf design. How many spine-to-leaf links are needed?

Target word: link count **Decides:** Each leaf connects to every spine.

A	Inverted operation	Adds instead of multiplying.
B	Correct	4×20 .
C	Dropped element	One link per leaf; drops the other spines.
D	Confusion set	Full mesh among leaves, not spine-leaf.

12. Answer C Concept 8 • transfer

A spine-leaf fabric has 96 spine-to-leaf links and 6 spines. How many leaves are there?

Target word: solve for leaves **Decides:** Links = spines $\times$ leaves.

A	Inverted operation	Subtracts instead of dividing.
B	Confusion set	Multiplies instead of dividing.
C	Correct	$96 / 6 = 16$.
D	Wrong stem number	The link count.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Control plane vs. data plane	The control plane decides where traffic should go (routing, policy); the data plane forwards packets. SDN centralizes the control plane in a controller.	Thinking the controller forwards user traffic itself.	Deciding paths vs. moving packets.
2	Northbound vs. southbound interfaces	Northbound APIs (often REST) connect applications to the controller; southbound protocols (OpenFlow, NETCONF, etc.) connect the controller to devices.	Calling OpenFlow a northbound API.	An app talking to the controller vs. the controller programming switches.
3	Intent-based networking	Operators state the desired business outcome; the system translates it into configuration and continuously verifies the network meets it.	Equating intent-based networking with scripting device commands.	A goal stated in business terms plus ongoing assurance.
4	Controller resilience	A single centralized controller is a single point of failure; deploy clustered controllers.	Assuming centralization makes the network automatically more reliable.	One controller managing the whole network.
5	Overlay vs. underlay	The underlay is the physical IP network; overlays (e.g., VXLAN) build virtual networks on top by encapsulating traffic.	Thinking an overlay replaces the physical network.	Virtual segments spanning a physical fabric.
6	Declarative vs. imperative automation	Declarative defines the desired end state and lets the system work out steps; imperative lists the exact steps to run.	Calling a step-by-step script declarative because it is automated.	“What it should look like” vs. “do this, then this”.
7	Idempotency	An idempotent automation task gives the same result whether run once or many times, changing only what differs from the desired state.	Assuming re-running automation always duplicates changes.	A playbook run a second time.
8	Spine-leaf fabric	Every leaf connects to every spine, so any two leaves are two hops apart and links = spines × leaves.	Connecting leaves to each other or counting links as spines + leaves.	Counts of spine and leaf switches.

Course scope (public description): Software Defined Networking (SDN) represents one of the fastest growing areas of network engineering. This course instructs learners on the SDN paradigm, which encompasses network automation, intent-based networking, and centralized network control. This course also teaches learners to view networking from a centralized and automated perspective rather than the traditional device-by-device model that is the legacy practice in many networks.